PRODUCT CONCEPT · FALL 2026

Pi Pals

A tiny computer, hiding inside a tiny friend.

Cute figurines with a Raspberry Pi Zero inside. Switch one on and it makes its own Wi-Fi network. Join it from your phone or laptop and you're in: a full shell, a desktop over VNC, and a battery level you can always see.



Concept brief · hardware, software & mockups

Working name; characters are original designs

THE IDEA

Something you'd carry around just because it's cute.

Most pocket computers look like bare circuit boards or grey boxes. A Pi Pal looks like a pumpkin on your desk, or a chick that sits on a keychain shelf, and it happens to be a working Linux computer you can reach from any device.

1

It makes its own Wi-Fi

Power on and within about 30 seconds the Pal shows up as a network, e.g. **Pip-7F3A**. You don't need a router, an app or any setup.

2

Your software, ready to go

Your shell and task manager come preinstalled. SSH into the shell, or VNC into a full desktop from a phone, tablet or laptop.

3

Always shows its battery

Rechargeable lithium cells inside. The battery % shows up in the shell prompt, the task manager and a little glowing heart.

Why it's lovable

People name their gadgets anyway. Giving the computer a face, a name and a personality ("Hi! I'm Pip.") turns a dev board into a companion. Seasonal characters make people want to collect them, and that's a reason to come back every few months.

The tech is cheap and already proven. A Pi Zero 2 W costs about $15, and hotspot, SSH and VNC are standard Raspberry Pi OS features. All the new work is in the character design, your software, and the battery integration.

Pocket coding lab
A real Linux box for students and tinkerers that fits in a backpack.

Desk buddy
Runs your scripts, bots and timers while sitting there looking adorable.

Offline file drop
Friends join its Wi-Fi to swap files at a café, a classroom or on a trip.

Gift & collectible
Seasonal drops: a pumpkin in October, a snowman in December.

> "Wait, the chick is a computer?"
>
> THE REACTION WE'RE DESIGNING FOR



Pip the Chick: the launch character

MEET THE LINEUP

One brain, many friends.

Every Pal uses the same internals and the same software image. Only the shell changes, so adding a new character means a new 3D model and a new name in the welcome banner. That's it.



Pip the Chick
Spring · Easter



Dot the Dino
All year



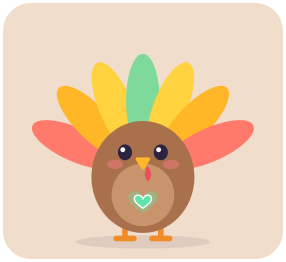
Penny the Piggy
All year · Gifts



Sunny
Summer



Gourdy the Pumpkin
Fall · Halloween



Gobbles the Turkey
Thanksgiving



Flurry the Snowman
Winter · Holidays



Hop the Bunny
Spring · Easter

Same heart in every Pal

Each character has a small heart on its chest that glows the battery colour. It's the one shared detail that makes the family recognisable.

Seasonal drops

Release a few characters each season: a bunny and chick for spring, the sun for summer, a pumpkin and turkey for fall, a snowman for winter. Dino and Piggy are the all-year staples.

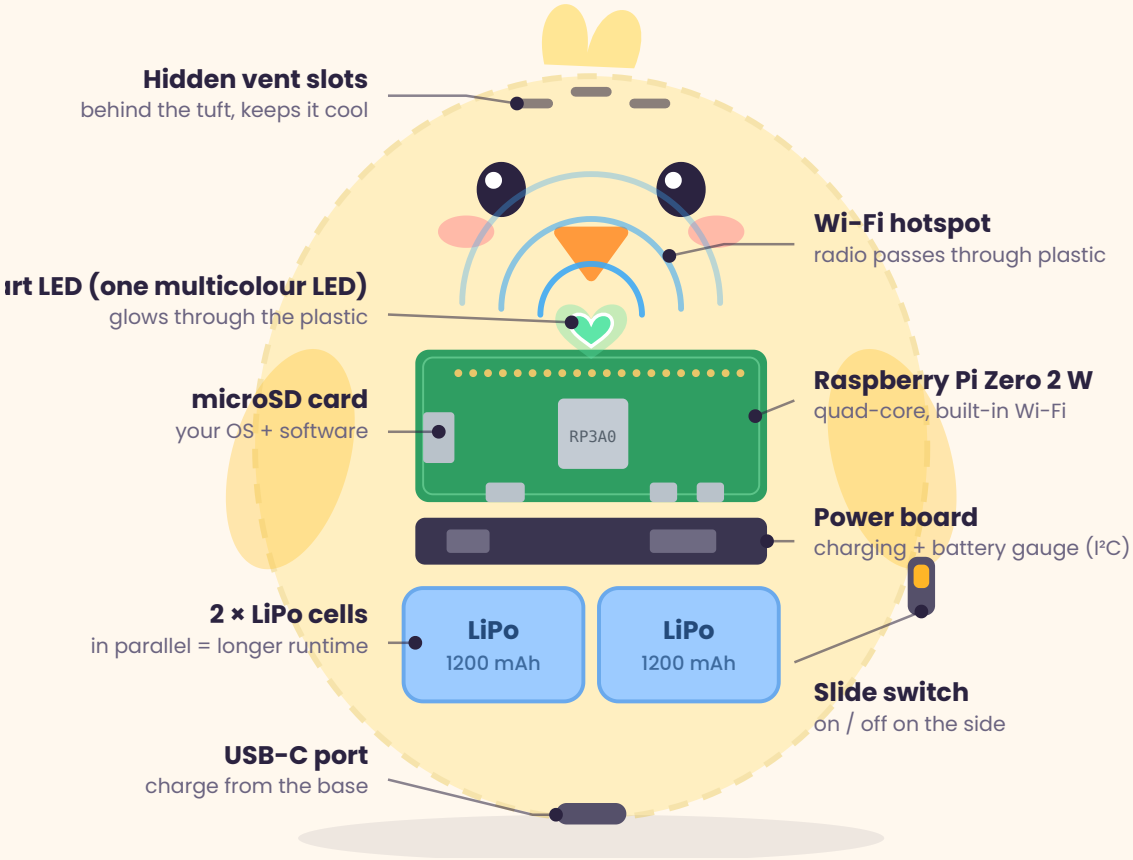
Personality in software

The hostname, Wi-Fi name, login greeting and prompt colours all match the character. `pip@pip`, `gourdy@gourdy`, and so on.

WHAT'S INSIDE

A cutaway of Pip.

Everything fits in a shell about the size of a large orange (roughly 90–100 mm tall, since the Pi board itself is 65 mm wide). The board sits up high so the Wi-Fi has a clear path out, and the batteries sit low so the Pal doesn't tip over.



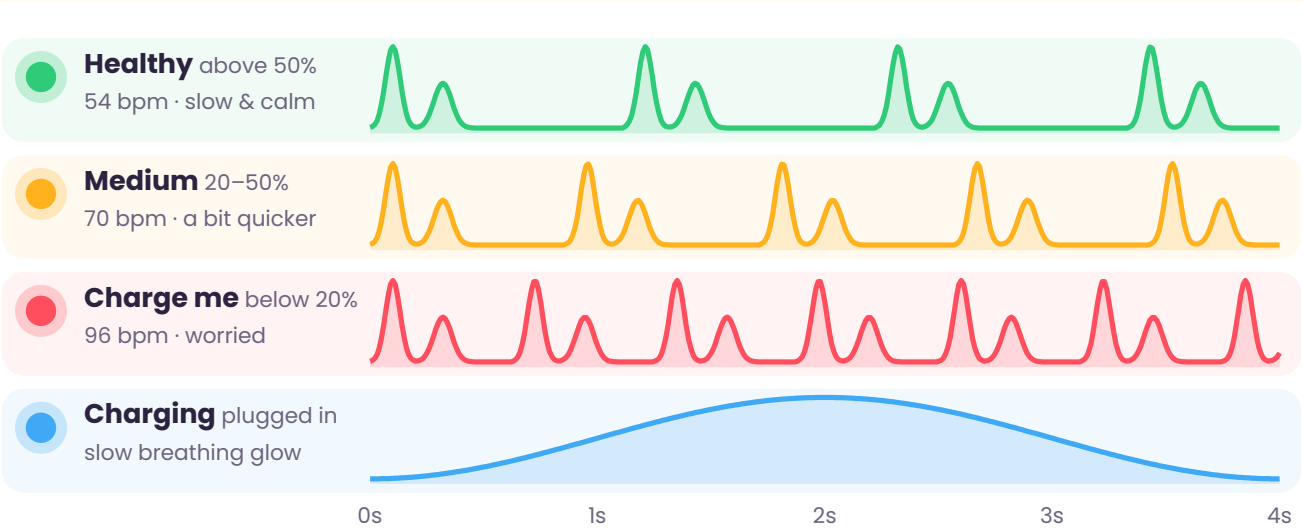
Part	Easy build	≈ Cost	Budget build	≈ Cost
Computer	Raspberry Pi Zero 2 W	$15	Raspberry Pi Zero 2 W	$15
Power & gauge	PiSugar board with battery (charging, switch, I²C gauge all in one)	$35–40	LiPo charger/boost board + MAX17048 fuel-gauge breakout	$5–8
Battery	Included (1200 mAh); add a 2nd cell for longer runtime	$0–8	2 × 1200 mAh protected LiPo, matched, in parallel	$10–14
Storage	16 GB microSD with your image	$4–6	16 GB microSD with your image	$4–6
Shell & bits	3D-printed shell, 1 smart RGB LED, screws	$3–5	3D-printed shell, 1 smart RGB LED, switch, USB-C jack, wire	$4–6
Per unit		**≈ $60–75**		**≈ $38–49**

Prices are rough retail single-unit estimates; check current prices before you plan around them. Buying in bulk brings the budget build down further. Use plastic or vinyl for the shell only, because metal or metallic paint blocks Wi-Fi.

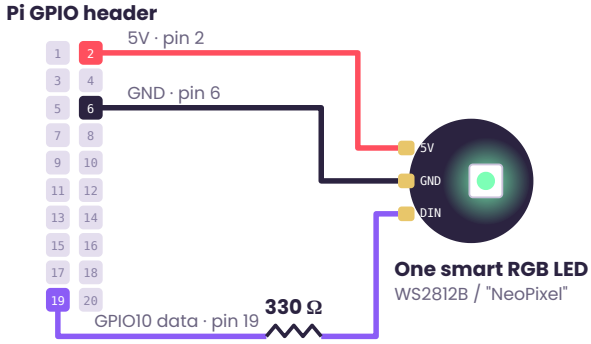
THE HEARTBEAT LIGHT

A heartbeat you can see.

A single multicolour LED on one GPIO pin shines onto the inside of a translucent heart, so the plastic itself glows. It pulses with a real lub-dub rhythm and never goes fully dark between beats. That's what makes the Pal feel alive.

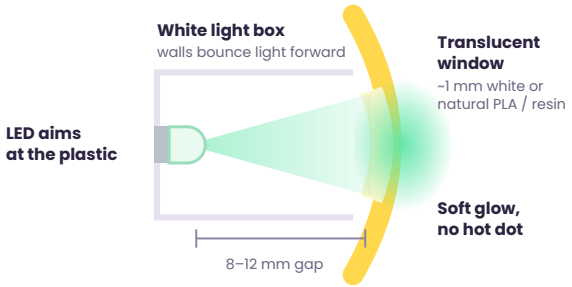


Wiring



Just three wires: power, ground and one data line. The LED has its own tiny chip that mixes any colour you ask for, so there's no resistor per colour. If it flickers, add a diode in the 5V wire.

Projecting through the plastic



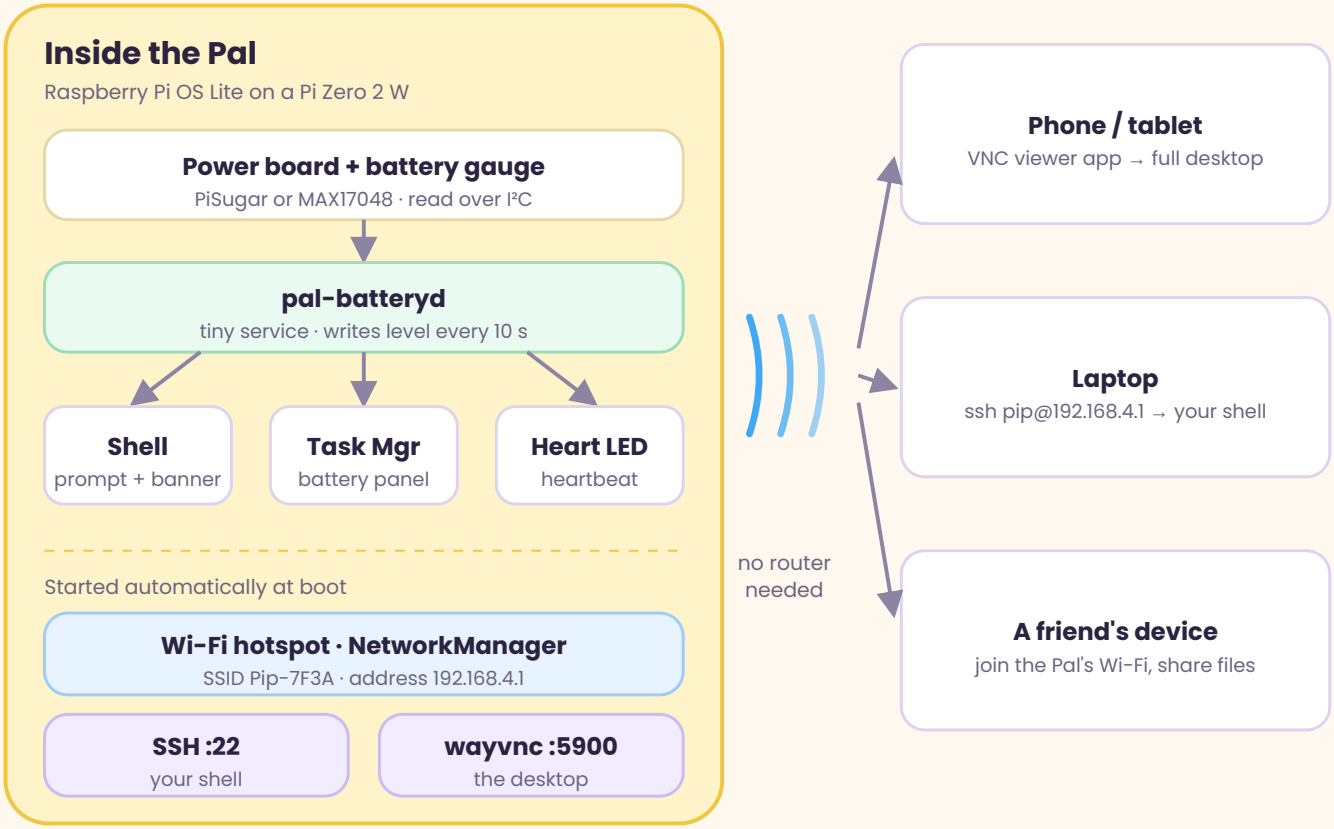
This is the same trick glowing gadgets use. Don't let the LED poke through. Aim it at a thin, frosted window from a short distance inside a small white box, and the light blooms into an even glow across the whole heart.

Battery	Colour	Heartbeat	What it says
Above 50%	Green	Slow lub-dub, ~54 bpm	"I'm happy and healthy."
20–50%	Yellow	A bit quicker, ~70 bpm	"Charge me sometime today."
Below 20%	Red	Fast, ~96 bpm; 120 bpm under 8%	"I need charging now."
Plugged in	Battery colour	Slow breathing, then steady green when full	"Recharging…"
pal wave	Rainbow	3-second colour sweep	"Here I am!"

HOW THE TECHNOLOGY FITS TOGETHER

Switch on, join, and you're in.

The Pal is a normal Raspberry Pi OS install with a few services that start at boot. One small battery service feeds the level to everything else, so the shell, the task manager and the LED always agree.



What happens when you flip the switch

0s — Power on
The heart blinks white while the Pi boots from the microSD.

15s — Battery service starts
Reads the gauge and the heart switches to the battery colour.

25s — Hotspot is up
The Pal's Wi-Fi appears with a unique password printed under its base.

30s — Ready
SSH and VNC are listening. Open a viewer app and the desktop is right there.

```
# hotspot: one-time setup, then automatic at every boot
nmcli con add type wifi ifname wlan0 con-name pal-ap \
  ssid Pip-7F3A autoconnect yes \
  802-11-wireless.mode ap ipv4.method shared \
  wifi-sec.key-mgmt wpa-psk wifi-sec.psk "$UNIT_PW"

# shell prompt shows battery from the battery service
PS1='\u@\h:\w [$(cat /run/pal/battery)%] \$ '
```

Built from proven pieces

- **NetworkManager** runs the access point, which is built into current Raspberry Pi OS.
- **wayvnc** shares the desktop, or a headless virtual one, over VNC.
- **pal-batteryd** is the only new service, about 50 lines of Python reading the gauge over I²C.
- A single **setup script** turns a fresh SD card into a Pal: it sets the name, hotspot, services and your software.

SOFTWARE MOCKUPS

What you see once you're connected.

The battery level shows up everywhere you look: in the welcome banner, in the prompt itself, and in a dedicated panel in the task manager.

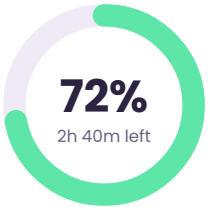
```
pip@pip: ~ (ssh 192.168.4.1)

   ( o>    Hi! I'm Pip the Chick.
   //\     Battery  ▰▰▰▰▰▰▰▰▱▱ 72%  ~2h 40m left · not charging
   V_/_    Wi-Fi    Pip-7F3A  · 2 friends connected
           Temp     47°C  ·  up 1h 12m

pip@pip:~ [72% ▼] $ pal battery
  Pack     3.92 V    2 × 1200 mAh in parallel (2400 mAh)
  Level    72 %      discharging at 310 mA
  Left     ~2 h 40 m at the current load
  Health   98 %      41 charge cycles
pip@pip:~ [72% ▼] $
```

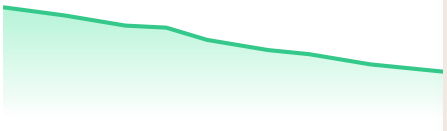
Task Manager: Pip

Processes | Performance | **Battery** | Network



Pack	3.92 V
Draw	310 mA
Charging	No
Cells	2 × 1200 mAh

Last 2 hours



Process	CPU	Battery impact
wayvnc	6.1%	High
NetworkManager	0.8%	Medium
your-app	2.4%	Low
pal-batteryd	0.1%	Low



Join the Pal's Wi-Fi, then open any VNC app

Above 50%
Green prompt. The heart beats slowly and calmly in green.

20–50%
Prompt and heart turn yellow, the beat quickens, and the banner suggests charging soon.

Below 20%
A fast red heartbeat, a warning goes to every open shell, and at 5% the Pal shuts down cleanly so nothing gets corrupted.

THE EXPERIENCE

From the box to the desktop in a minute.

Nothing to install and nothing to configure. A small card tucked under the base has the Wi-Fi name, the password and a QR code that joins the network in one scan.

1



Flip the switch

The switch is on the side of the Pal. The heart blinks while it wakes up.

2



Scan the card

The QR code joins `Gourdy-2C1B`. Each unit has its own password.

3



Open your shell or VNC

Use SSH from a laptop or any VNC viewer app on a phone or tablet.

4



Charge over USB-C

Plug in under the base. The heart breathes slowly while it charges, then glows steady green when full.

How long will it run?

Estimates for a Pi Zero 2 W that's broadcasting Wi-Fi. Real numbers depend on what your software is doing, so measure them on the prototype.

What it's doing	1 cell 1200 mAh	2 cells 2400 mAh
Hotspot on, sitting idle	~5 h	~10 h
Someone in the shell	~4 h	~8 h
VNC desktop in use	~2.5–3 h	~5–6 h

Assumes about 0.7–1.4 W draw and roughly 85% boost-converter efficiency. Always wire matched, protected cells in parallel, never in series.

Small touches that add charm

- **A greeting with personality.** Each Pal says hi in its own voice: Gourdy is a little spooky, Flurry is chilly, Sunny is cheerful.
- **The heart is the status light.** A slow green heartbeat means healthy, yellow means medium and a fast red beat means it needs a charge. A slow breathing glow means it's charging.
- **A "find my Pal" command.** Run `pal wave` from your shell and the heart flashes rainbow.
- **A friends counter.** The banner shows how many devices are connected right now.
- **A sleepy mode.** After 30 idle minutes Wi-Fi power-saving kicks in to stretch the battery.

MAKING & SELLING IT

Very attainable. Here's a path.

Every part is off the shelf, so you can start with one working chick and grow from there. The main things to plan for are the battery, radio rules and how the product is labeled.

Roadmap

Step 1 — **One working Pip**
Easy build with a PiSugar and a 3D-printed chick. Prove the hotspot, VNC and battery-in-the-shell flow, then measure real runtime and heat.

Step 2 — **The Pal image**
A setup script that turns a blank SD card into any character: name, Wi-Fi, greeting, colours and your software.

Step 3 — **Small batch of 10**
Switch to the budget build, refine the shell so it snaps together with screws, and give units to friends for feedback.

Step 4 — **Seasonal drops**
Launch with Pip, Dot and Gourdy, then add a new character each season.

Possible price points

Starter	Plus	Limited
$59	**$79**	**$99+**
1 cell, all-year characters	2 cells, any character	seasonal, numbered run

Placeholders to test against your costs and what buyers respond to; they aren't market research.

Things to plan for

- **Shipping lithium.** Batteries packed inside equipment have carrier rules (UN3481), so check them before you sell by mail.
- **Radio rules.** The Pi Zero 2 W's Wi-Fi is pre-certified, but a finished product may still need FCC/CE steps. Check before selling widely.
- **It's a gadget, not a toy.** Label it for ages 14 and up. Marketing to young children brings toy-safety testing.
- **Security.** Give every unit its own Wi-Fi and VNC password. Never ship a shared default.
- **Heat & Wi-Fi.** Hide the vents in the design and keep the shell non-metallic.

